



Carboxymethyl tamarind kernel gum bio-hydrogel for soil amendment and micronutrient release in sustainable agricultural technologies

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Abstract

Water scarcity poses significant challenges to agricultural productivity, leading to substantial crop losses each year. In this study, we synthesized a superabsorbent, eco-friendly bio-hydrogel from carboxymethyl tamarind kernel gum (CMTKG), using epichlorohydrin as a crosslinking agent. The hydrogel is designed to act both as a water reservoir and a slow-release carrier for essential micronutrients. The bio-hydrogel synthesized under optimized conditions exhibited a swelling of 23,845% and a reswelling capacity of 6,217% after five cycles at room temperature (25 °C). The swollen hydrogel retained water in air for up to 16 days. When incorporated into soil, this retention period was extended by about 48 days compared to soil without the hydrogel. The maximum water-holding capacity of the soil was increased by 90% compared to the control soil sample. Biodegradation analysis showed a weight loss of the hydrogel by 98% after 60 days. In a blackberry plantation study, the germination rate and survival rate were 25% and 85% higher, respectively, than those planted in the control soil. As a proof of the concept, molybdenum (Mo) was loaded into the CMTKG-based bio-hydrogel to demonstrate controlled release of micronutrients. The hydrogel released approximately 86% of the loaded Mo into water over a period of 19 days. Kinetic modeling showed that both the first order and Peppas–Sahlin models provided the best fit, indicating that Mo release kinetics involve a combination of concentration-gradient-driven transport and CMTKG polymer chain relaxation dynamics.

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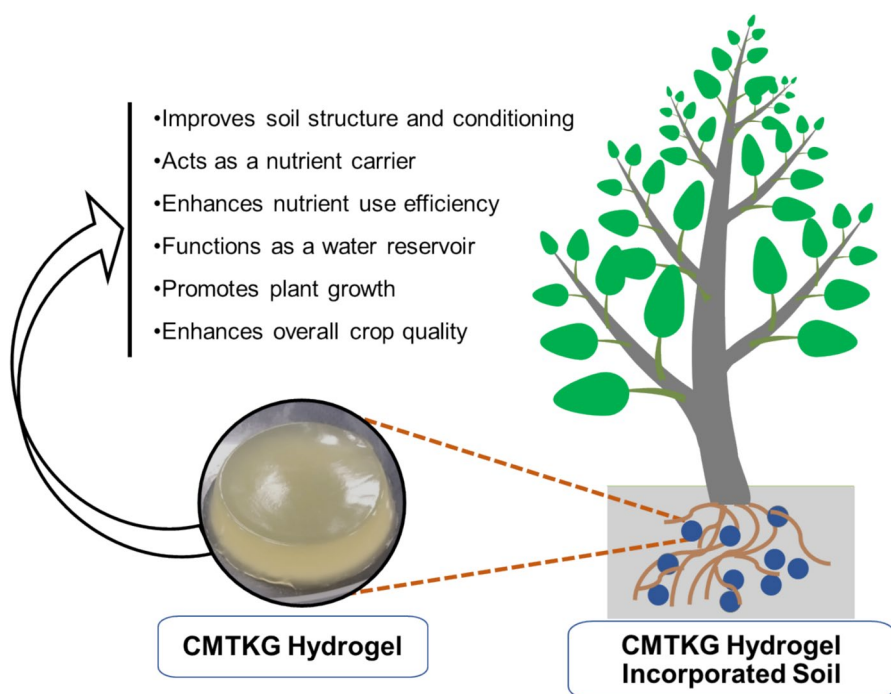
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Graphical abstract



Carboxymethyl tamarind kernel gum (CMTKG) as a water reservoirs and plant micronutrients carriers for sustained release

Keywords Bio-hydrogel · Micronutrient · Molybdenum · Slow release · Soil amendment

Introduction

Hydrogels are a type of three-dimensionally cross-linked polymer with a highly hydrophilic nature that can absorb and retain huge amounts of water even under harsh atmospheric conditions such as heat, pressure, and salinity [1]. Hydrogels are produced from synthetic polymers or biopolymers. Synthetic polymers are harmful to the human body and the environment, but biopolymers are non-toxic and eco-friendly [2] So, scientists are more interested in biopolymer-based hydrogels for their biodegradation, which is beneficial to agriculture and the environment [3]. At present, superabsorbent bio-hydrogel is attracting attention from researchers for its remarkable water retention properties, fertilizer/nutrient carrier, biocompatibility, high stability, tunability, and biodegradation [4].

Bio-hydrogels can be used as soil conditioning to improve the soil water management and plant micronutrient release, making them especially worthy in arid or

drought-prone regions [5]. These bio-hydrogels can retain large amounts of water and release it slowly over time, ensuring plants accessible to moisture. These boost the reduction of irrigation frequency and enhance healthy plant growth, seed germination, root growth, microbial activities, and crop yield, particularly in arid-prone areas. These bio-hydrogels enrich the physical, mechanical, and fertility characteristics of soil [6].

Although synthetic polymers dominate hydrogel production polymers [7, 8], nature-derived biopolymers such as cellulose, starch [9, 10], chitosan [11] and alginate [12, 13] bio-polymers are increasingly employed in bio-hydrogels for carrying nutrients or fertilizers because they are readily available, environmentally friendly, and inexpensive. Carboxymethyl cellulose and poly(4-vinylpyridine) were used to load and release the urea [14]. Hydroxyl propyl methyl cellulose and polyvinyl alcohol were used to load and release the urea [15]. Biochar-based controlled-release nitrogen fertilizers were fabricated using urea-loaded biochar, bentonite, and polyvinyl alcohol [16].

Conventional farming fails to generate sustainable agricultural production in extreme environmental conditions, such as drought, temperature, and salinity [17]. Also, conventionally used fertilizers are inefficient in reaching the target because of the excessive loss of fertilizers; only 20–30% of the used fertilizer can be absorbed, and the remaining portion is lost through nitrification, volatilization, and leaching [18]. So, micronutrients and fertilizers-loaded hydrogels are more supportive and can play a vital role in controlled and optimum sustained release. Therefore, the hydrogel-based slow or controlled release systems are helpful and could release an optimum amount of water and agrochemicals for a sustained period [19].

This research study aims to synthesize Carboxymethyl Tamarind Kernel Gum (CMTKG)-based bio-hydrogel with high water absorption and water retention and nutrient carrier vehicle. CMTKG is a biopolymer derived from Tamarind Kernel gum (TKG) extracted from *Tamarindus indica* L seeds. This polysaccharide is made up of glucose, xylose, and galactose in the ratio of 3:2:1. It has a linear β -(1,4)-D glucose backbone with partial α -D-xylo-pyranose (C-1) substitution at the O glucopyranosyl (C-6) residues, which are further substituted at C-2 by β -D galactopyranosyl units (C-1). TKG and alkaline mono-chloroacetic acid by Williamson's etherification (S_N^2 mechanism) make CMTKG, having carboxymethyl groups ($-\text{CH}_2\text{COOH}$) at its C-6 position. The anionic sodium salt of CMTKG is applied to synthesize bio-hydrogel. CMTKG has a long life span, more swelling power, and high solubility in cold water and is stable in most solvents compared to TKG [20]. So, water and nutrient retention and release can be improved by using CMTKG-based bio-polymer.

Several biopolymer-based hydrogels and their combinations were studied for water absorption, water retention, and slow/controlled micronutrient/fertilizer release. Carboxymethyl cellulose/chitosan hydrogel was concocted for water retention, showing a swelling capability of 70–110 g/g [21]. Jackfruit peel cellulose nanocrystal–alginate hydrogel showed a swelling ratio of more than 300% [22]. Carboxylated cellulose nanofibrils (CCNFs) and carboxymethyl cellulose-g-poly (acrylic acid-co-acrylamide) were synthesized, exhibiting swelling capacity enhancement from 245.8 to 458.7 g/g with increasing CCNF content [23]. Cellulose

nanofibers and carboxymethyl cellulose-based hydrogels were loaded with urea, showing 33% released in 10 days in DI water and a weight loss of about 81% after 90 days [24]. Sodium carboxymethyl cellulose/hydroxyethyl cellulose blend filled with 5% of spherical regenerated cellulose composite hydrogels were synthesized to study the monoammonium phosphate (MAP) fertilizer release of 12 h in water and 19–20 days in soil as compared to 1–2 h in water and 9–10 days in soil of uncoated MAP [25]. Nano-cellulose/sodium alginate/metal–organic frameworks (MOFs) hydrogels were fabricated for urea release, showing that only 50% of the urea is released on the 30th day [26].

In this study, Mo was loaded in CMTKG bio-hydrogels as a proof-of-concept measurement. Mo is an essential micronutrient for plants, as it contributes to the nitrogen-fixing enzyme nitrogenase [27]. Very small amounts of Mo are required for plants [28]. It plays a vital role in nitrate reductase and aldehyde oxidase in the enzymes of plants [29]. Some *Brassica* crops like cauliflower and rapeseed are greatly affected in production by the deficiencies of Mo [30]. High rainfall soil area often suffers from Mo deficiencies [31]. Acid weathered soils and well-drained soils are vulnerable to Mo retention [29]. Foliar application of Mo enhances bean yield ~4.3-fold in Brazil [28]. Mo enriched seeds enhanced cowpea yield by 42–70% compared to normal seeds. It enhances plant growth with more nodules and leaf area [28]. Its deficiencies hamper wheat growth and reduce nutritional quality in the grain [32]. In this case, soil conditioning with Mo-loaded bio-hydrogel can be a good option to supply water as well as Mo in the soil.

The objective of this study was to synthesize CMTKG bio-hydrogel using epichlorohydrin (ECH) as a crosslinker. The prepared bio-hydrogel was characterized using Fourier transform infrared spectroscopy (FTIR), scanning electron microscopy (SEM), thermal gravimetric analysis (TGA), rheological analysis, and UV–visible spectrophotometer analysis. The developed bio-hydrogel was analyzed regarding its swelling, deswelling, water retention, recyclability, and biodegradation. The impact of bio-hydrogels was observed by the growth of Indian blackberry (*Syzygium cumini* L.). Furthermore, a micronutrient Mo was loaded in CMTKG bio-hydrogel and release parameters were investigated with different kinetic models.

Experimental

Materials and instrumentation

Carboxymethyl tamarind kernel gum (CMTKG) with 0.20° of substitution ($MW = 8.5 \times 10^7$ g/mol) was received from Hindustan Gum and Chemicals Ltd., Bhiwani, Haryana, India. It was isolated from the seeds of the *Tamarindus Indica* L. plant with the molar ratios of xylose, galactose, and glucose being 1:2:3. This is a natural polymer that is also among the least expensive gums in the market. Epichlorohydrin (ECH) (98%), hydrochloric acid (36%), sodium hydroxide (98%), pyrazine-2,3-dicarboxylic acid (98%), and molybdic acid (99%) were purchased from Thermo Fisher Scientific (Delhi, India) and were used as received. Deionized (DI) water was made in our laboratory. All chemicals were used as received.

A magnetic stirrer with temperature-controlled hot plate (Remi Elektrotechnik Limited, New Delhi, India) was used. A hot-air oven (PSO—451, Presto Stantest Private Limited, Haryana, India) was used to dry polymers and hydrogels. An FTIR spectrophotometer (PerkinElmer spectrum version 10.5.3) was used to gather IR spectra of the CMTKG and hydrogels. Surface morphologies were analyzed by scanning electron microscopy (SEM) (JEOL JSM-6610LV). For SEM imaging, all samples were dried overnight in a hot air oven at 60 °C. The samples were then coated with gold via sputtering, using a gold sputter coater JEC 300. The base pressure in the SEM chamber during analysis was 10^{-7} Torr. Thermogravimetric analysis (TGA) was performed by PerkinElmer TGA 4000 under a nitrogen atmosphere, and the temperature range was from 20 to 900 °C with a 10 °C /min heating rate. MCR-101 rheometer (Anton Paar, Japan) with a parallel plate was used to perform the rheological properties. The concentration of released molybdenum from hydrogels was measured using a UV–Vis SHIMADZU UV-900I spectrophotometer (Tokyo, Japan). A calibration curve was created by measuring known concentrations of molybdenum, and this curve was used to determine the unknown concentrations of the molybdenum solution.

Preparation of CMTKG hydrogel

CMTKG and NaOH were added to DI water and stirred for 1.5 h to form a homogeneous mixture (Table 1). Then, ECH was added to the solution and stirred for 30 min. The solution was then placed in a hot air oven at 60 °C for 5 h to complete the gelation process (Fig. 1a). When ECH is introduced to CMTKG, the hydroxyl groups on CMTKG get attacked by the epoxide ring, creating a covalent link between CMTKG and epichlorohydrin through the formation of an ether bond (Fig. 1b).

Swelling study

The dried CMTKG hydrogel was weighed (W_o) and then immersed in the solution for 72 h at room temperature (~25 °C). Then, the swollen hydrogel was removed from the solution and weighed (W_E). The weighing process was repeated until a small decrease in weight was observed. The swelling ratio (SR) was calculated using the formula specified below [33].

$$\text{SR (\%)} = \frac{W_E - W_o}{W_o} \times 100\% \quad (1)$$

Recyclability experiment

The recyclability (reswelling–deswelling) of the CMTKG hydrogel was assessed by immersing hydrogel samples of known weights in sufficient DI water for 72 h to reach equilibrium swelling. Subsequently, the swollen hydrogel samples

Table 1 Formulations of CMTKG hydrogels with swelling ratio (SR) and reswelling ratio (Re-SR)

Code	CMTKG (g)	ECH (mL)	SR (%)	1st Re- SR (%)	2nd Re- SR (%)	3rd Re-SR (%)	4th Re-SR (%)	5th Re-SR (%)
C-1	0.1	1	Disintegrative hydrogel					
C-2	0.2	1						
C-3	0.3	1		18,804	18,005	17,865	8587	5087
C-4	0.4	1		23,845	23,713	23,011	12,302	6217
C-5	0.5	1		22,181	21,570	20,985	11,583	6158
C-6	0.6	1		21,058	20,415	19,987	11,023	6005
C-7	0.3	1.5		18,376	18,178	17,987	9534	6025
C-8	0.4	1.5		22,321	21,189	20,884	9875	6085
C-9	0.5	1.5		21,008	20,949	20,076	8978	5889
C-10	0.6	1.5		19,524	19,002	18,896	7895	5589
C-11	0.3	2		18,066	17,525	16,125	6589	5486
C-12	0.4	2		20,576	19,987	19,897	7328	5514
C-13	0.5	2		19,288	18,895	18,156	5143	4986
C-14	0.6	2		18,558	17,567	17,132	4987	4125

SR (%), Re-SR (%) were measured at room temperature ($\sim 25^{\circ}\text{C}$), NaOH (0.7 g) and DI water (10 mL) were kept constant across all the hydrogel formulations

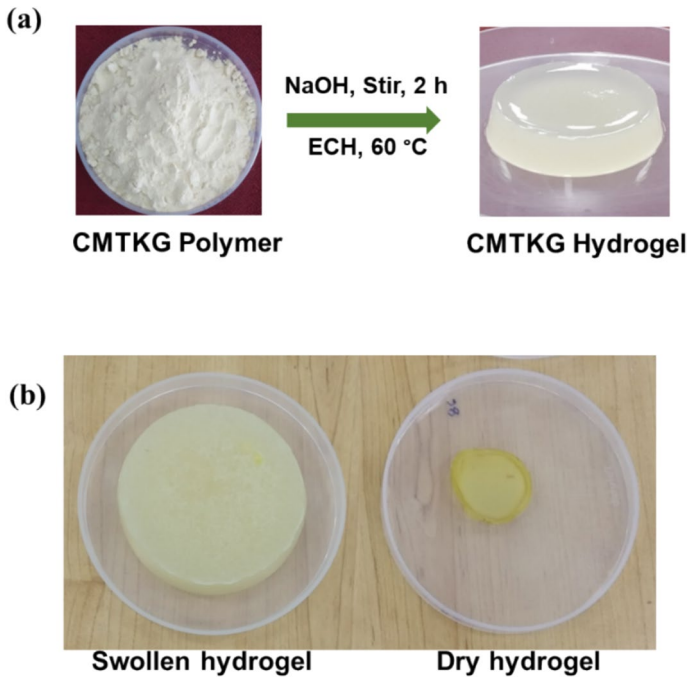


Fig. 1 **a** Schematic diagram of synthesis of CMTKG hydrogel and **b** Swollen and dry hydrogel,

were removed from the solution and weighed. Then, the swollen hydrogels were oven-dried and retrieved. The swelling of the dried hydrogel sample was measured again, following a similar procedure for a second cycle. This process was repeated 5 times to evaluate the samples' recycling potential. The reswelling (%) was calculated according to Eq. 1.

Water retention in air

The oven-dried CMTKG hydrogel was submerged in DI water for 72 h. The swollen hydrogels (W_o) were taken out of the water and kept at room temperature ($\sim 25^\circ\text{C}$) (relative humidity $\sim 63\%$), and their weight was recorded over time. The water retention of the CMTKG hydrogel sample in air was measured from the following equation[34, 35].

$$\text{Water retention(\%)} = \frac{W_t}{W_o} \times 100\% \quad (2)$$

Here W_t is the weight of the sample at time (t).

Water retention in soil/hydrogel mixture

Oven-dried soil (100 g) was mixed with oven-dried CMTKG hydrogel (0.1 g) and placed in a plastic pot. Similarly, a control soil sample was prepared. 100 mL of water was poured into the pots, and each experimental setup was maintained same environment (room temperature ~ 25 °C and $\sim 63\%$ relative humidity). The weights of the pots were measured every day until no change in weights was observed. The water retention of the soil/CMTKG-hydrogel mixture and control soil was calculated using the equation specified below [36]:

$$\text{Water retention(\%)} = \frac{H_t}{H_o} \times 100\% \quad (3)$$

where H_t is the weight of water present in the soil at a time, t , and H_o is the initial weight of water in the soil.

Maximum water holding capacity (MWHC)

Maximum water holding capacity for hydrogels in agriculture is crucial as it can contribute soil water management. With retention of large amounts of water, it can help plants survive droughts and promote healthier plants for the enhancement of production [37]. The MWHC of the soil was calculated by placing a fixed quantity of soil in a beaker with small holes at the bottom, and a filter paper was kept at the bottom to restrain soil loss. The soil/hydrogel mixture beaker and the control soil sample beaker were placed beneath a water tap to allow maximum water absorption. Removing excess water, the net weight of water absorption in control soil and soil/hydrogel mixture was measured from the weights of the wet soil, dry soil, and filter paper using the following formula [38]:

$$\text{MWHC} = [\text{Wet soil} - \text{Dry soil}] - [\text{Wet filter paper}] \quad (4)$$

We have measured moisture of the water holding capacity by weighing the pots [38].

Biodegradation studies

Biodegradable hydrogels are more important in agriculture and for the environment. These eco-friendly biodegradable hydrogels can boost soil moisture and fertility, seed germination, and crop production and also aid in controlled, slow release of micronutrients or fertilizer, and sustainable agricultural farming practices [39]. So, the biodegradation of the CMTKG hydrogels was studied over 60 days using a soil burial method [38]. Pre-weighed CMTKG hydrogel samples (2 g) were buried in well-moisturized soil at a 6–7 cm depth within a nylon sieve bag. The CMTKG hydrogel samples were retrieved at specific intervals (7, 15, 30, and 60 days). We have maintained moisture by regularly monitoring with adjustment of the water

content in soil for biodegradation study. For these purposes, we have added 100 mL DI water once in a week (at room temperature $\sim 25^\circ\text{C}$, relative humidity $\sim 63\%$). Then, it was washed carefully to remove any attached soil, dried in a hot air oven, and reweighed to determine the biodegradation.

The residual dry content (RDC) and weight loss (%) were measured using the following equations [40].

$$\text{RDC (\%)} = \frac{D_t}{D_o} \times 100\% \quad (5)$$

$$\text{Weight loss(\%)} = \frac{D_o - D_t}{D_o} \times 100\% \quad (6)$$

where D_o is the undegraded dry weight of the hydrogel, and D_t is the dry weight of the degraded hydrogel.

Pot plantation study

The performance of CMTKG hydrogel was studied by observing the impact of the hydrogel on the germination and growth of Indian blackberries (*Syzygium cumini* L.) in a pot experiment. The plantation studies were conducted using three pots containing 5% (w/w) CMTKG hydrogel, 1% (w/w) CMTKG hydrogel, and a control (without hydrogel). 15 seeds were sown in each pot, 3.0 cm below the soil surface. All pots were placed in the same environment with the same amount of water. The CMTKG hydrogel was buried at a depth of 5–6 cm below the soil surface. We have monitored seed germination and germination rate, i.e., germination energy. It

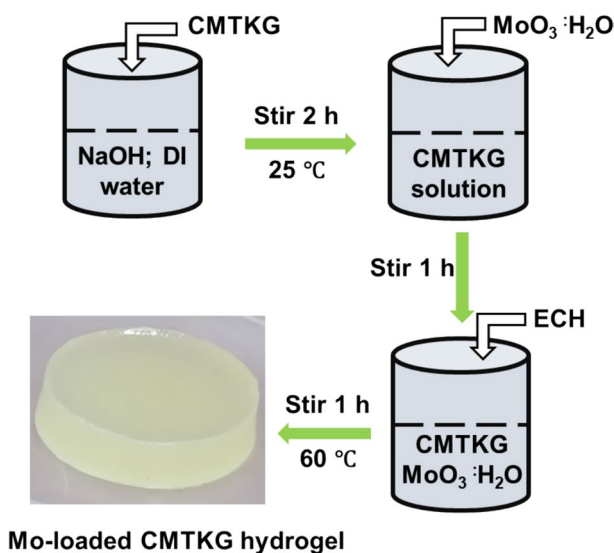


Fig. 2 Schematic diagram of synthesis of Mo-loaded CMTKG hydrogel

determines how quickly a batch of seeds germinate under optimal conditions, indicating their vigor. A higher germination energy suggests a faster and more uniform emergence of seedlings. Seed germination energy was calculated by using the following equation [14, 17].

$$\text{Germination energy(\%)} = \frac{S_n}{T_n} \times 100 \quad (7)$$

where S_n is the total number of seeds that have germinated on the n^{th} day following sowing, and T_n is the total number of seeds sown.

After germination, the blackberry plants were allowed to grow for 90 days and watered on the 90th day. Thereafter, no water was given to the plants for the next 30 days. The plants' survival was observed in these 30 days. This study was conducted in the laboratory, DTU between October 2024 and March 2025. The survival rate was calculated based on the following equation [17].

$$\text{Survival(\%)} = \left(1 - \frac{W_n}{T_n} \right) \times 100 \quad (8)$$

where W_n is the number of plants that were wilted on the n^{th} day.

Synthesis of Mo-loaded CMTKG hydrogel

CMTKG (0.4 g), NaOH (0.7 g) and DI water (10 mL) were taken to a beaker and stirred for 2 h at room temperature ($\sim 25^\circ\text{C}$). A homogeneous suspension was formed, and 0.01 M molybdic acid was added and stirred 1 h continuously. Then, 1.0 mL ECH was added to the mixture and stirred 1 h at room temperature ($\sim 25^\circ\text{C}$). The resultant mixture was placed in a hot air oven at 60°C for 6 h to form Mo-loaded CMTKG hydrogel (Fig. 2).

Loading efficiency of molybdenum

The loading efficiency of molybdenum into CMTKG hydrogel is given below.

$$\text{Loading efficiency(\%)} = \frac{H_{\text{Mo}} - H_o}{H_o} \times 100\% \quad (9)$$

where H_{Mo} is dry weight of Mo-loaded CMTKG hydrogel, and H_o is dry weight of CMTKG hydrogel.

Molybdenum release in water and release kinetics

For molybdenum release study in water, 100 mg Mo-loaded CMTKG hydrogel was submerged in 100 mL DI water, and then 5 mL Mo released water was withdrawn at 24 h time interval and continued up to 20 days. The releasing media was replenished

with DI water for volume unchanged. The cumulative release (%) of molybdenum was calculated by the following equation [40]:

$$R_{\text{cum}}(\%) = \frac{M_t}{M_{\infty}} \times 100 \quad (10)$$

where M_t is the amount of molybdenum released at a time (t), and M_{∞} is the amount of molybdenum released at equilibrium.

For Mo concentration measurement, 5 mL of Mo released sample solution was mixed with 35 mL DI water and 10 mL HCl. Then, 1 mL SnCl_2 and 2.5 mL pyrazine-2,3-dicarboxylic acid were added and heated at 60 °C for 5 min. The solution was diluted with DI water, and the dilution was performed in a 100-mL volumetric flask. The absorbance of the solution at 525 nm was determined, and Mo concentration was detected using a standard calibration curve [42].

The molybdenum release kinetics were studied with the following models (Table 2) [43]:

Statistical analysis

We performed three identical experiments for each study. Errors were calculated based on these experiments and included where appropriate. We have also added statistical analyses for the pot plantation on blackberry. All data were analyzed by one-way ANOVA ($p=0.05$).

Results and discussion

Characterization of CMTKG polymer and hydrogel

FTIR spectra of CMTKG polymer and CMTKG hydrogel were recorded to characterize the chemical compounds of the biopolymers and hydrogels (Fig. 3a). Broad peaks between 3400 and 3216 cm^{-1} are attributed to the O–H stretching vibration [44]. C–H stretching vibrations were observed between 2952 and 2850 cm^{-1} [45]. The strong peak at 1035 cm^{-1} is attributed to C–O, C–C, and C–O–C stretching vibrations. Peak at 1750 cm^{-1} represents unsubstituted –COOH. Peak at 1577 cm^{-1} represents COO^- asymmetric vibration [38]. Peak at 1409 cm^{-1} represents CH_2 stretching vibration.

SEM images of CMTKG and CMTKG hydrogel were recorded to observe the three-dimensional views of surfaces and morphology. SEM image of CMTKG polymer shows a granular structure (Fig. 3b). It is a compact and non-porous surface. The surface also looks like a coil. CMTKG bio-hydrogel exhibits a porous and spongy rough surface (Fig. 3c). The pore diameter was calculated in the range of 40–75 μm . Some interspatial voids also visible in CMTKG bio-hydrogels [38].

Thermogravimetric analysis (TGA) was performed to observe the thermal stability, decomposition, durability and potential environmental impact. The weight loss of polymer and hydrogel was clearly visible as shown in Fig. 4. The first

Table 2 Different kinetics models with explanations

Kinetics models	Equation	Explain
Korsmeyer–Peppas model	$\frac{M_t}{M_\infty} = K_{kp} t^n$	K_{kp} and n are release constant and diffusion exponent, respectively. When $n < 0.45$, the process follows the Fickian diffusion. In the range $0.45 < n < 0.89$, the diffusion process is non-Fickian or anomalous release. If $n > 0.89$, the process is dominated by super case II transport
Higuchi model	$\frac{M_t}{M_\infty} = K_H t^{\frac{1}{2}}$	K_H is the Higuchi dissolution constant. It follows the Fickian diffusion
Peppas–Sahlin model	$\frac{M_t}{M_\infty} = k_1 t^m + k_2 t^{2m}$	k_1 and k_2 are diffusional constants related to Fickian and non-Fickian mechanism contribution. m is the coefficient of the purely Fickian diffusion exponent
Zero-order kinetics model	$\frac{M_t}{M_\infty} = K_o t$	K_o is the release constant of zero order. The rate of diffusing species released is constant and independent of its concentration
First-order kinetics model	$\frac{M_t}{M_\infty} = 1 - e^{-K_1 t}$	K_1 is the release constant. The rate of diffusing species released is linearly dependent on the concentration of the diffusing species

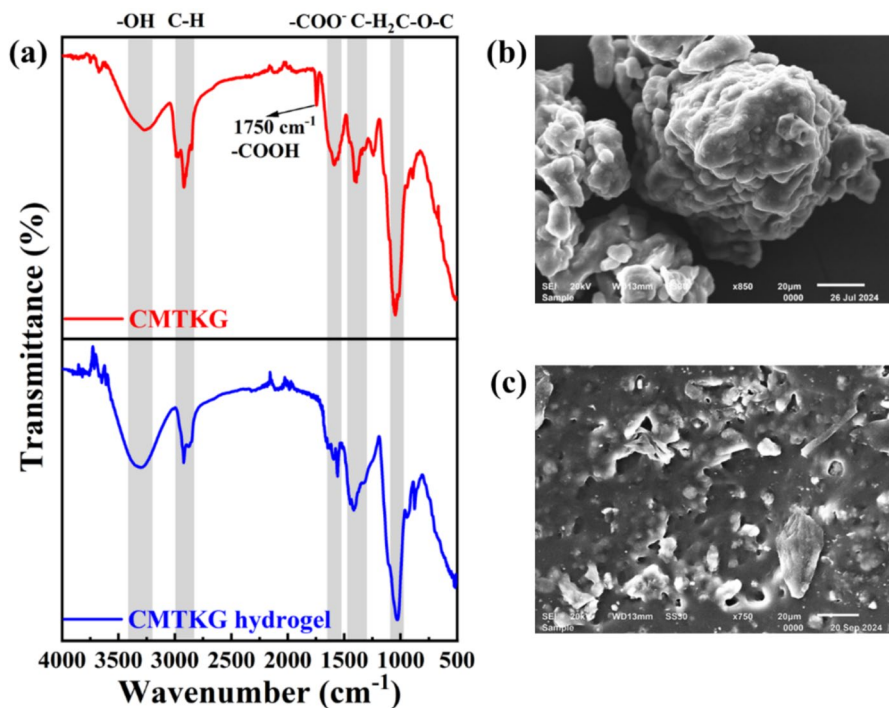


Fig. 3 **a** FTIR spectra of CMTKG and CMTKG hydrogel; SEM images of **b** CMTKG, **c** CMTKG hydrogel (Hydrogel formulation C-4)

weight loss of CMTKG polymer and CMTKG hydrogel was observed as 11.5 and 4.8%, respectively, due to the removal of moisture. The second weight loss could be due to the degradation of -OH moieties and -COOH moieties in both polymer and hydrogel. The third and subsequent weight loss could be due to the degradation of the polymeric backbone of CMTKG and CMTKG hydrogel [38]. More weight loss was observed in CMTKG polymer compared to CMTKG hydrogel due to the formation of H-bonding and crosslinking in the hydrogel. Although the onset temperature of CMTKG polymer (255 °C) is larger than CMTKG hydrogel (235 °C), the overall thermal stability is possessed by CMTKG hydrogel. At 900 °C, the total weight loss of CMTKG and CMTKG hydrogel was observed as 86.7 and 62.7%, respectively.

The rheological evaluation for the amplitude sweeps test and angular frequency test was performed to observe the viscoelastic behavior of CMTKG hydrogel (C-4–6). The CMTKG hydrogel showed the viscoelastic region between 0.5 and 10.1% of strain (Fig. 5a–c). The maximum values of the storage modulus (G') and loss modulus (G'') of CMTKG (C-6) hydrogel were observed at 41.85 and 15.44 Pa, respectively, and the cross-over point was found at around 10.1% shear strain (Fig. 5c), which indicates physical changes from solid to liquid [46]. The C-6 gave the highest modulus and crossover point due to the highest CMTKG concentration.

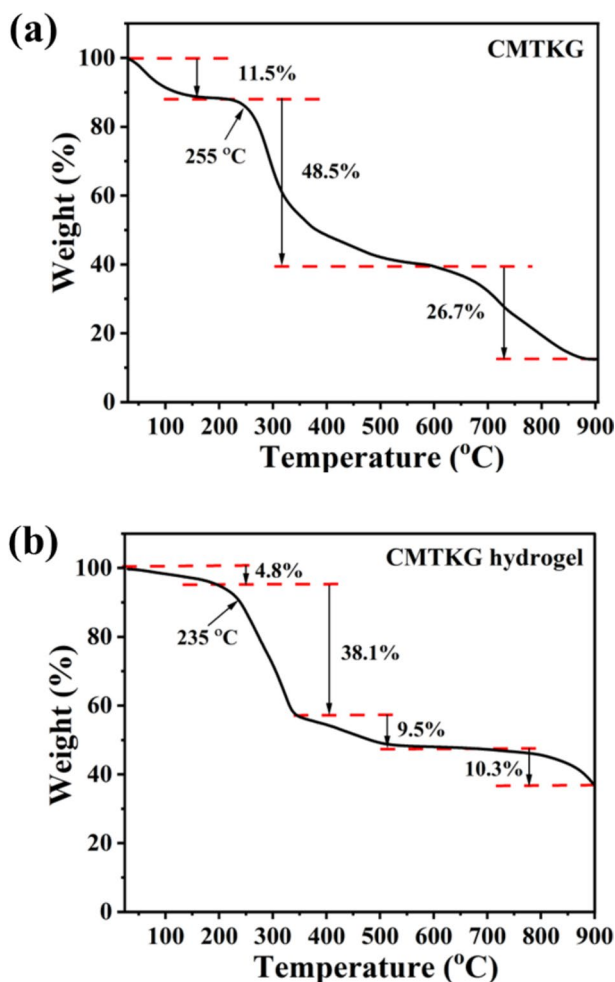


Fig. 4 TGA spectra of **a** CMTKG and **b** CMTKG hydrogel. (Hydrogel formulation C-4)

Angular frequency sweep test of CMTKG hydrogel (C-4–6) was performed (Fig. 5d–f), which indicates that the storage modulus was higher than the loss modulus. So, these hydrogels are elastic. The storage modulus, loss modulus, and crossover point of C-6 were higher than C-4 and C-5 due to CMTKG concentration.

In all CMTKG hydrogels, storage modulus (G') predominates over loss modulus (G'') up to the crossover point, revealing that elastic (solid-like) behavior prevails over viscous (liquid-like) behavior, indicating a typical gel-like behavior [47]. The viscous behavior predominates ($G'' > G'$) after the crossover point of all CMTKG hydrogels and viscoelastic moduli decrease with strain amplitude, resulting in high deformation with structural breakdown [48].

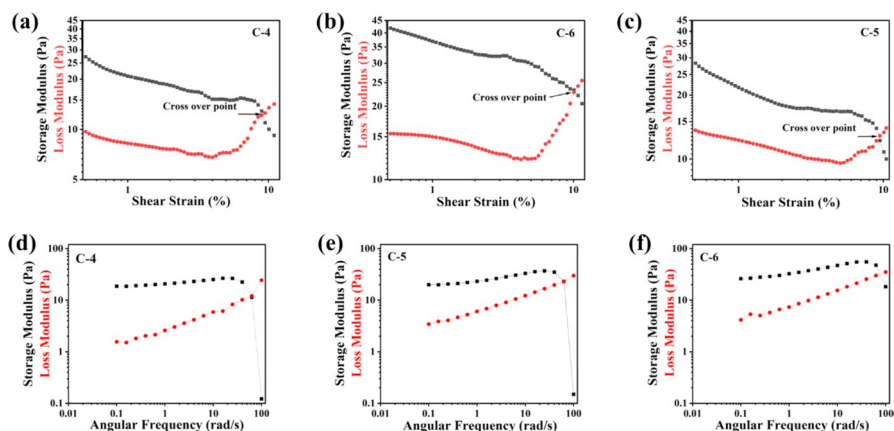


Fig. 5 Storage modulus and loss modulus as a function of shear strain and angular frequency **a, d** C-4 **b, e** C-5 **c, f** C-6

Swelling behavior

The swelling capacity of CMTKG hydrogel was evaluated by the variation of the concentration of polymer and crosslinking agent. The optimized CMTKG hydrogel (C-4) was used to evaluate the swelling in different temperatures, saline solutions, and pH.

Effect of biopolymer and crosslinker

The effect of biopolymer concentration on swelling is shown in Fig. 6a. The swelling of CMTKG hydrogel increases with the enhancement of the

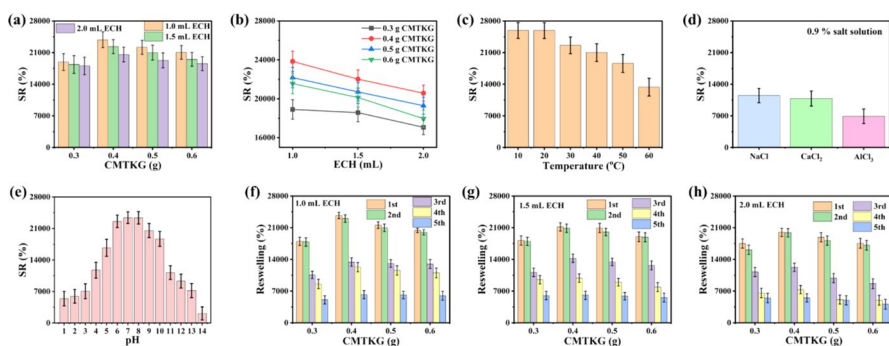


Fig. 6 **a** Effect of CMTKG concentration on SR at different ECH concentrations, **b** Effect of ECH concentration on SR at different CMTKG concentrations, **c** Effect of temperature on SR, **d** Effect of saline solutions on SR, **e** Effect of pH on SR, **f** Reswelling vs CMTKG concentration for 1.0 mL ECH, **g** Reswelling vs CMTKG concentration for 1.5 mL ECH, **h** Reswelling vs CMTKG concentration for 2.0 mL ECH

concentration of CMTKG. Swelling enhances with the concentration of CMTKG to a particular point and then decreases when the concentration of CMTKG exceeds 0.5 g in 10 mL DI water. As CMTKG contains $-\text{COOH}$ groups, more CMTKG creates more $-\text{COOH}$ groups, making an augmented repulsion among the $-\text{COOH}$ groups. It reinforces the chain relaxation and creates more space to accommodate water molecules. As a result, swelling increases with the enhancement of the concentration of CMTKG [38]. A decrease in swelling was observed as the concentration of CMTKG increased to 0.5 g/10 mL.

The effect of concentration of crosslinking agent, ECH on swelling for CMTKG hydrogel is shown in Fig. 6b. It follows Flory–Rehner theory [49]. The swelling decreases with the increases in concentration of ECH from 1.0 to 2.0 mL. More crosslinking networks are formed with the enhancement of concentration of ECH. As a result, pore spaces among the network decrease and lessen the free volume inside the network of CMTKG hydrogel and diminishing the water absorption [49].

Effects of temperature, salts solution, and pH

The swelling of CMTKG hydrogel was studied in DI water at different temperature (10–60 °C) as shown in Fig. 6c. This temperature range was selected for swelling of CMTKG hydrogel as in agricultural production, temperature varies in different region, season and time. The swelling decreases with increasing temperature of 10 to 60 °C. At higher temperature H-bonds in CMTKG hydrogel are broken, resulting in contraction and thus decrease in water absorption [34, 50].

The swelling behavior of CMTKG hydrogel was studied with different saline (0.9% NaCl, 0.9% CaCl_2 , and 0.9% AlCl_3) solutions (Fig. 6d) at room temperature (~ 25 °C). It was observed that SR (%) is lower in saline solution compared to DI water, because of ionic strength affected the swelling [51]. The ionic strength of the cations affects the osmotic pressure between the outer and inner medium of the CMTKG hydrogel and thus hindering the absorption of water molecule into the hydrogel. Higher ionic charge creates the resistive forces to diffuse water molecules into hydrogels, and therefore, swelling decreases profoundly. The swelling of CMTKG hydrogel in ionic saline solution was followed the sequence $\text{NaCl} > \text{CaCl}_2 > \text{AlCl}_3$. For charge neutralization, more COO^- groups are required for trivalent (Al^{3+}), compared to bivalent (Ca^{2+}) and monovalent (Na^+) cations.

The swelling behavior of CMTKG hydrogel was studied at different pH (1–14) solution (Fig. 6e). Various pH solutions were prepared by adding HCl and NaOH. The swelling increases initially with the enhancement of pH from 1 to 7 and a decrease in swelling occurred with increasing pH from 8 to 14. The $-\text{COOH}$ groups in CMTKG hydrogel exist as $-\text{COO}^-$ ions at pH 7, repelling neighboring groups and more space is created and allowing more water absorption in CMTKG hydrogel [26]. When HCl or NaOH was added to the solution for pH adjustment, it reduced the charge density of the $-\text{COO}^-$ groups; as a result, the swelling decreased.

Recyclability studies

The recyclability of CMTKG bio-hydrogel was examined for 5 cycles of reswelling at room temperature (25 °C) as shown in Fig. 6f–h. All bio-hydrogels exhibited good reswelling capacity, with the C4 hydrogel (0.4 g CMTKG) demonstrating superior reswelling compared to other CMTKG formulations. The amount of crosslinking agent (ECH) also influenced the reswelling, hydrogels formed with 1.0 mL ECH reswelled more than those with 1.5 mL or 2.0 mL ECH. The good recyclability of CMTKG bio-hydrogel will be useful for agricultural applications. After 5 cycles of deswelling and swelling, all the bio-hydrogels showed a downward trend. The decrease of SR (%) after 5 cycles may be due to the disintegration of internal physical crosslinking points during deswelling and swelling [52].

Water retention in air and soil/hydrogel mixture

The water retention of CMTKG bio-hydrogel was evaluated by keeping the swollen hydrogel in air at room temperature (~25 °C). The CMTKG hydrogel can retain water for ~16 days (Fig. 7a). Water retention in soil is very important for plants. The CMTKG bio-hydrogels can absorb and release water for plant growth in drought-prone regions [34].

The water retention of CMTKG bio-hydrogel was studied in a soil/hydrogel mixture (Fig. 7b). Bio-hydrogel boosts water retention in soil. Water evaporated rapidly in the control soil, but slowly evaporated in the soil/bio-hydrogel mixture. The CMTKG bio-hydrogel/soil mixture retained water for 67 days, whereas the control soil sample retained water for 19 days. These findings confirmed the water retention capability of CMTKG bio-hydrogel. The mixture of CMTKG bio-hydrogel into the soil delayed the evaporation of water from the soil. So, these CMTKG bio-hydrogels boost the plant growth in arid regions [53].

Maximum water holding capacity

Hydrogels with the maximum water holding capacity improve water retention capacity in soil. It reduces water wastage and promotes water retention in the soil. Figure 7c shows the MWHC for soil/hydrogel mixture. Control soil sample retained 33.2 g of water per 100 g of soil. When hydrogels were incorporated in soil

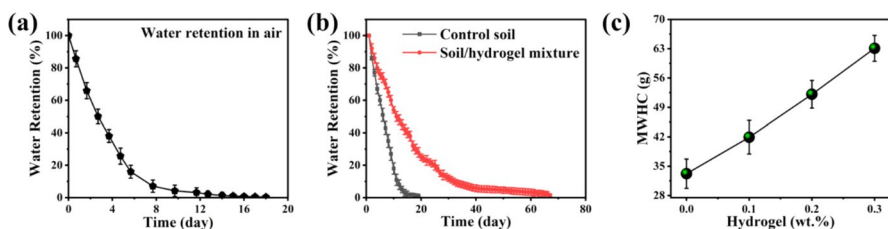


Fig. 7 a Water retention of CMTKG hydrogel in open air over time b water retention of CMTKG hydrogel/soil mixture over time c MWHC of soil at different hydrogel amounts

at concentrations of 0.1%, 0.2%, and 0.3%, the MWHC was increased by 41.9 g, 52.1 g, and 63.1 g, respectively. The MWHC increased linearly with the enhancement of the hydrogel in soil. Therefore, CMTKG hydrogel enhanced water retention in soil as it retained water.

Biodegradation studies

The biodegradation of CMTKG hydrogel was studied over 60 days using a soil burial method. The biodegradation rate of CMTKG hydrogel was impetuous at the initial stage, followed by a sluggish biodegradation (Fig. 8a). After 30 days, the weight

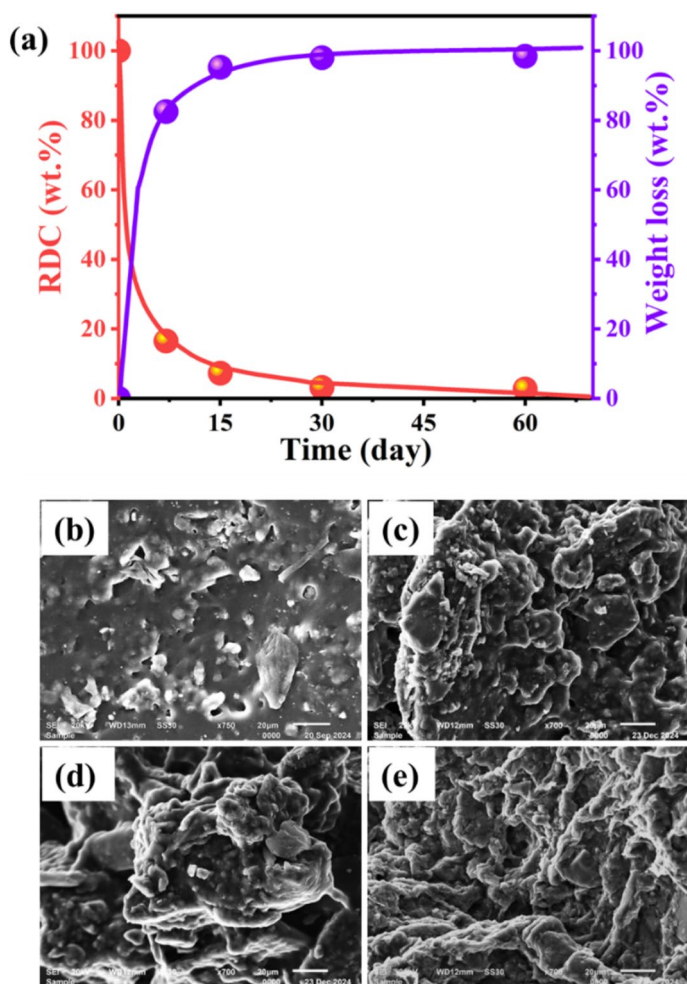


Fig. 8 Biodegradation of the hydrogels in soil: **a** Residual dry content (RDC) and weight loss over time. SEM images of hydrogels at **b** day-0, **c** day-15, **d** day-30, and **e** day-60 of decomposition (Hydrogel formulation C-4)

loss and RDC reached a steady state. The weight loss of CMTKG hydrogel was determined as 82.5%, 95.3%, 97.9%, and 98.4% by 7, 15, 30, and 60 days, respectively. The residual dry content of CMTKG hydrogel was measured as 16.5%, 7.3%, 3.1%, and 2.7% by 7, 15, 30, and 60 days, respectively. The CMTKG bio-hydrogel degrades rapidly due to water absorption, swelling, and microorganism activities [38]. The high level of microbial activity in the early stage results in rapid degradation [54]. The SEM images of CMTKG hydrogels showed degradation compared to non-degraded sample (Fig. 8b–e) [49].

Pot plantation study

The CMTKG hydrogel was evaluated on blackberry plant growth under water stress conditions (Fig. 9a–c). It was observed that the plant in 5% hydrogel-treated soil grew faster compared to the 1% hydrogel-treated soil. The germination rate was monitored for 6 weeks, revealing that 5% of hydrogel-treated soil demonstrated high germination compared to 1% of hydrogel-treated soil and control soil (Fig. 9d). The plant height and leaf number were monitored up to 12 weeks, indicating that hydrogel-treated soil surpassed the control soil (Fig. 9e–f). After 13 weeks, no water was supplied to the plants, and the survival of the plants was observed. After 4 weeks, the control soil plants exhibited complete wilting; however, the plants cultivated in hydrogel-treated soil remained alive and healthy. (Fig. 9c, g). The dry shoot and root weight were measured, revealing that the plant in the hydrogel-treated soil prevailed over the control soil (Fig. 9h).

Mo-loaded CMTKG hydrogel and Mo release kinetics

Agricultural soils need micronutrients to enhance crop production. Slow and controlled release micronutrients and fertilizers are useful in agriculture to extend the release of micronutrients and fertilizers into the soil. The structure of hydrogels

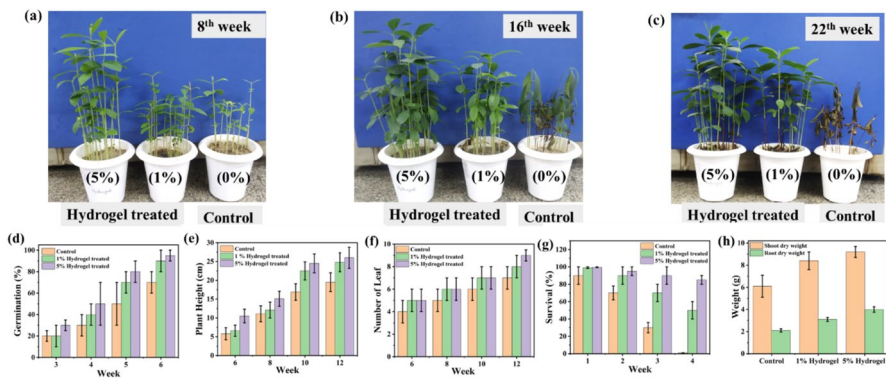


Fig. 9 Blackberry pot plantation **a** after 8 weeks, **b** after 16 weeks, **c** after 22 weeks, **d** Germination vs time **e** Plant height vs time, **f** Number of leaf vs time **g** Survival rate vs time, and **h** Shoot and root dry weight

allows for the loading and subsequent slow or controlled release of micronutrients and fertilizers. Consequently, we chose molybdenum (Mo), an essential plant micro-nutrient, as a proof-of-concept measurement.

FTIR spectra of CMTKG hydrogel, molybdic acid and Mo-loaded CMTKG hydrogel were investigated to confirm the existence of Mo in the hydrogel (Fig. 10a). The FTIR spectra of molybdic acid show a broad peak at $\sim 3155\text{ cm}^{-1}$, indicating -OH stretching vibration [55]. A peak at 1400 cm^{-1} indicates O–H–O bending vibration [55]. The peak at 873 cm^{-1} illustrates the Mo=O, Mo–O, Mo–O–Mo stretching/bending vibration in molybdic acid [56, 57]. In Mo-loaded CMTKG hydrogel, a characteristic peak of Mo=O stretching vibration at 933 cm^{-1} is visible. The SEM image of Mo-loaded CMTKG hydrogel is shown in Fig. 10b. The SEM image of the Mo-loaded CMTKG hydrogel reveals some visible cracks and the presence of stone-like structures of varying sizes.

The cumulative release (%) of Mo from CMTKG hydrogel was studied (Fig. 10c), indicating that the cumulative release (%) was more than 80% after 11 days. Five kinetic models were studied to observe the release behavior of Mo in water (Fig. 10c). It is observed that the first-order model ($R^2=0.98$) and the Peppas–Sahlin ($R^2=0.97$) model are the best fit models (Table 3). Mo release from CMTKG bio-hydrogel is concentration dependent, and the release rate is directly proportional to the concentration of Mo remaining in the CMTKG bio-hydrogel. The release process also dominates the kinetics through diffusion and relaxation of the polymer.

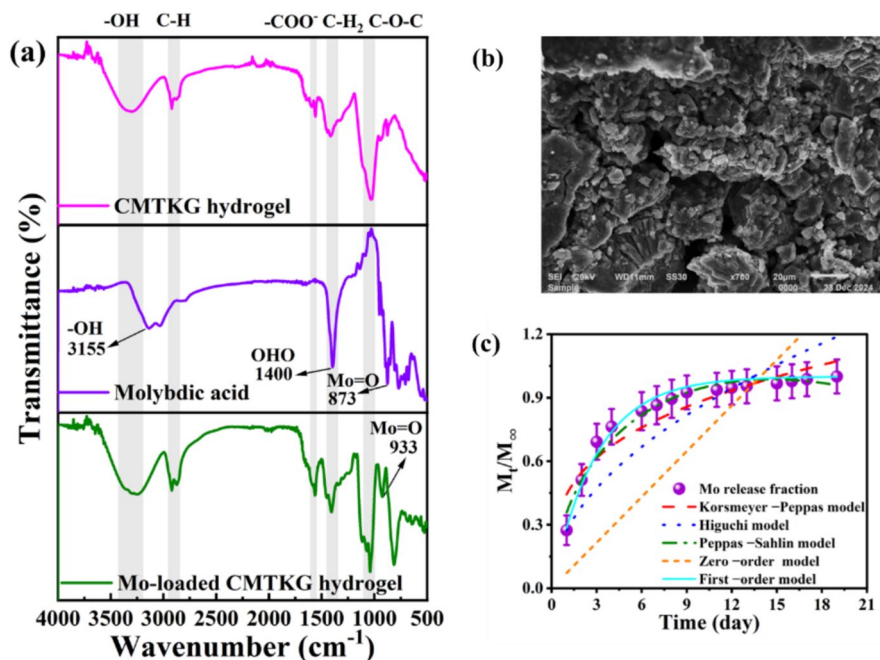


Fig. 10 **a** FTIR spectra of CMTKG hydrogel, molybdic acid, and Mo-loaded CMTKG hydrogel; **b** SEM image of Mo-loaded CMTKG hydrogel; **c** Mo release kinetics of different models

Table 3 The molybdenum releases kinetic parameters for CMTKG hydrogel samples

Korsmeyer–Peppas model			Higuchi model		Peppas–Sahlin model				Zero-order model		First-order model	
K_{KP}	n	R^2	K_H	R^2	k_1	k_2	m	R^2	K_0	R^2	K_1	R^2
0.44	0.3	0.89	0.27	0.65	0.4	−0.04	0.6	0.97	0.07	0.38	0.34	0.98

In Peppas–Sahlin model, the first kinetic constant ($k_1=0.4$) showed the Fickian diffusion, which dominates the mechanism of Mo release from CMTKG hydrogel and the second kinetic constant ($k_2=-0.04$) indicated polymer relaxation, As, k_2 is smaller than k_1 , so, diffusion mechanism is surpassed to relaxation for the release of Mo from CMTKG hydrogel. The diffusional exponent ($m=0.6$) demonstrates that the release of Mo from CMTKG hydrogel is the combination of both diffusion and relaxation. A comparative study was performed to analyze the swelling ratios, biodegradation rates, water retention and nutrient release performance (Table 4).

Statistical analysis of blackberry plant growth parameters [application of 1 wt% CMTKG hydrogel and 5 wt% CMTKG hydrogels]

One-way ANOVA was analyzed for the sum of squares (SS), degree of freedom (df), and mean squares (MS) between the germination, plant height, number of leaf and survival of blackberries; 1 wt% CMTKG hydrogel and 5 wt% CMTKG hydrogels were applied for blackberries growth. The F test and p value were evaluated for growth parameter of blackberries. The p values in the F test corresponding to germination (Table 5), plant height (Table 6), leaf number (Table 7) and survival (Table 8) of blackberries are 0.578704, 0.655637, 0.421875 and 0.211567, respectively, which is not significant at 5% (0.05) level of significance.

Conclusion

Slow release of water and micronutrients using bio-hydrogels is a sustainable method to reduce the adverse effects on the environment. Hence, we have developed an environmentally friendly, cost-effective bio-hydrogel based on CMTKG crosslinked with ECH for a sustainable agricultural application. The prepared hydrogel was characterized by FTIR, SEM, TGA, UV–visible, and rheological techniques to observe the hydrogel formation and stability. The swelling and absorption of the superabsorbent bio-hydrogels were assessed in DI water, at different temperatures, pH, and in saline solutions. The highest swelling was obtained at 23845% in DI water (Formulation 0.4 g CMTKG, 0.7 g NaOH, 1.0 mL ECH, and 10 mL DI water) at room temperature ($\sim 25^\circ\text{C}$). The water retention of the bio-hydrogel/soil mixture and control soil showed 67 days and 19 days, respectively. When 0.1%, 0.2%, and 0.3% CMTKG bio-hydrogels were mixed with soil, the maximum water-holding capacity of the soil was increased by $\sim 26\%$, $\sim 57\%$, and $\sim 90\%$, respectively

Table 4 The swelling ratios, biodegradation rates, water retention and nutrient release performance were compared to other articles

Composition of hydrogel	Swelling ratios, biodegradation rates, and water retention	References
Amla, acrylic acid (AA), Acrylamide (AAM), N, N'-methylene-bis-acrylamide (MBA), ammonium persulfate (APS)	Swelling capacity 900%, biodegradation in burial and vermicompost soil within 72 and 64 days, water retention 6 days, potassium release 943.97 after 750 h	[58]
Amla, AA, AAM, MBA, APS	Swelling capacity 300%, biodegraded by 95.6% over 25 days, water retention 7 h, release nutrient 692.14 ppm after 70 h	[59]
Amla, AA, AAM, MBA, APS	Swelling ratios 520%, after 36 days, hydrogel exhibited 88% degradation in vermicompost soil and 81% degradation in burial soil	[60]
Carboxymethyl tamarind kernel gum (CMTKG), NaOH, urea, epichlorohydrin (ECH)	Swelling capacity 15,359%, retained water in the soil for up to 56 days, urea release 95% after 8 days in water and 34 days in soil	[40]
Cellulose, AA, ECH, potassium persulfate (KPS)	Swelling capacity 9811%, water retention 4 days, biodegradation 10% after 120 days	[41]
Sodium alginate (SA), KPS, 2-acrylamide-2-methyl-1-propane sulfonic acid, N-isopropylacrylamide, humic acid	Swelling ratio 40,000%, water retention 144 h, Urea release 94% after 4 days	[34]
CMTKG, AA, AAM, MBA, KPS	Swelling capacity 189%, water retention 43%, biodegradation ~43% after 60 days	[20]
Cellulose, urea, NaOH, ECH	Swelling capacity 220%, urea loading capacity 1100%	[46]
Agar, orange peel, limonene	Swelling capacity 650 to 700% after 4 h, biodegradation 52% after 90 days	[61]
CMTKG, AA, NaOH, MBA, KPS	Swelling capacity 648%, water retention 58%, biodegradation 41% after 20 days	[38]
CMTKG, NaOH, ECH	Swelling capacity 23,845% in DI water, water retention 67 days, biodegradation 98% after 60 days, molybdenum release ~81% after 11 days	This study

Table 5 One-way ANOVA for the responses (average germination of blackberries)

	SS	df	MS	F	<i>p</i> value
Between Groups	600	2	300	0.6	0.578704
Within Groups	3000	6	500		
Total	3600	8			

Table 6 One-way ANOVA for the responses (average plant height of blackberries)

	SS	df	MS	F	<i>p</i> value
Between Groups	44.41	2	22.20	0.453294	0.655637
Within Groups	293.89	6	48.98		
Total	338.3	8			

Table 7 One-way ANOVA for the responses (average number of leaf of blackberries)

	SS	df	MS	F	<i>p</i> value
Between Groups	2	2	1	1	0.421875
Within Groups	1	6	1		
Total	3	8			

Table 8 One-way ANOVA for the responses (survival of blackberries)

	SS	df	MS	F	<i>p</i> value
Between Groups	1596.169	2	798.0844	2.034684	0.211567
Within Groups	2353.44	6	392.24		
Total	3949.609	8			

compared to the control soil sample. The biodegradability of the formulated bio-hydrogels was confirmed by the soil burial method, which resulted in a mere 2.7% residual dry content after 60 days. The bio-hydrogels proved effective in supporting plant growth. Specifically, the germination rate was 25% higher, and the survival rate was 85% higher in the bio-hydrogel-treated soils compared to the control soil. As a proof-of-concept measurement, a micronutrient, Mo, was loaded in CMTKG bio-hydrogel, showing a cumulative release of ~81% in DI water after 11 days. The first-order model and the Peppas–Sahlin model were the best-fit models, indicating that Mo release is concentration dependent with Fickian diffusion and polymer chain relaxation.

Author contributions Tarun Kumar Gayen: Data curation, Formal analysis, Investigation, Validation, Writing—original draft, Visualisation, Conceptualization, Methodology, Writing—review & editing. Mohammad Amdad Ali: Supervision, Investigation, Writing—review & editing. Sudhir G Warkar: Supervision, Investigation, Writing—review & editing.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interests The authors declare no competing interests.

Ethics approval and consent to participate This manuscript is an original paper and has not been published in other journals. The authors agreed to keep the copyright rule.

Consent for publication The authors revised the manuscript and agreed with the publication.

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